PROJECT DESIGN PHASE

1. PHASE OVERVIEW

The Project Design Phase focuses on designing the structure and workflow of the Auto Ticket Classification using Flow Designer project.

Based on the requirements identified in the previous phase, the system is designed to provide a structured ticket management process in ServiceNow.

The design includes the custom ticket table, required fields, Category and Subcategory structure, dependent choices, Flow Designer automation, ticket classification logic, and email notification process.

2. SYSTEM DESIGN

The proposed system is designed using ServiceNow and Flow Designer.

The overall system consists of the following major components:

1. Custom Ticket Management Table
2. Ticket Fields
3. Category and Subcategory Configuration
4. Category–Subcategory Dependency
5. Flow Designer Automation
6. Ticket Classification Logic
7. Email Notification

These components work together to automatically classify IT support tickets.

3. SYSTEM WORKFLOW DESIGN

The overall workflow of the system is designed as follows:

User Creates IT Support Ticket

↓

Ticket Record Created

↓

Flow Designer Trigger

↓

Check Short Description

↓

Identify Issue Keyword

↓

Assign Category

↓

Assign Subcategory

↓

Update Ticket

↓

Send Email Notification

↓

Final Ticket Output

The workflow is designed to reduce manual classification and provide consistent ticket processing.

4. TICKET TABLE DESIGN

A custom table named Custom Incident Workflow is designed to store the IT support ticket information.

The table is designed with the following fields:

Field | Type / Purpose

Number | Auto Number

Caller | Reference

Category | Choice

Subcategory | Dependent Choice

Short Description | String

Description | String

State | Choice

Assigned Group | Reference

Assigned To | Reference

The source project document specifies these fields as part of the custom ticket table design.

5. CATEGORY DESIGN

The Category field is designed to represent the major types of IT support issues.

The following Categories are included:

Network

Hardware

Access

Performance

Each Category is associated with a corresponding Subcategory.

6. **SUBCATEGORY DESIGN**

The Subcategory field is designed as a dependent choice field.

The following Subcategories are included:

Wi-Fi

Projector

Forgot Password

Slow Computer

The Subcategory values are designed to depend on the selected Category.

7. CATEGORY–SUBCATEGORY DEPENDENCY DESIGN

The dependency relationship is designed as follows:

Category | Subcategory

Network | Wi-Fi

Hardware | Projector

Access | Forgot Password

Performance | Slow Computer

This design ensures that the Subcategory is logically related to the selected Category.

The project document specifically requires the Subcategory field to depend on Category.

8. TICKET CLASSIFICATION DESIGN

The automatic classification logic is designed using predefined keywords in the Short Description.

The classification structure is:

Keyword / Issue | Category | Subcategory

Wi-Fi / Network | Network | Wi-Fi

Projector | Hardware | Projector

Forgot Password | Access | Forgot Password

Slow Computer | Performance | Slow Computer

The Flow Designer uses these conditions to determine the appropriate classification.

9. FLOW DESIGNER STRUCTURE

The Flow Designer is designed to automate the ticket classification process.

The flow contains:

Trigger

Record Created

The flow starts when a new ticket record is created.

Condition

The flow checks whether the Category field is empty before applying the classification logic.

Classification Conditions

The flow checks the Short Description for the required issue keywords.

The main conditions include:

Wi-Fi

Projector

Password

Slow Computer

Actions

Based on the identified issue, the flow performs an Update Record action to assign the required Category and Subcategory.

The project source describes the Flow Designer structure using these conditions and Update Record actions.

10. EMAIL NOTIFICATION DESIGN

An automated email notification is included in the system design.

The notification is designed to inform the caller after the ticket is created.

The email contains a confirmation message related to the submitted IT support request.

The Flow Designer uses a Send Email action for this purpose.

11. COMPLETE AUTOMATION DESIGN

The complete automation is designed as:

Record Created

↓

Category is Empty

↓

Check Short Description

↓

Wi-Fi Projector Password Slow Computer

↓

Network Hardware Access Performance

↓

Wi-Fi Projector Forgot Password Slow Computer

↓

Update Record

↓

Send Email

This design represents the complete automated classification workflow.

12. USER INTERFACE DESIGN

The ticket form is designed to allow users to enter the required information.

The main user-facing fields include:

Caller

Category

Subcategory

Short Description

Description

State

Assigned Group

Assigned To

The design provides a structured form for creating and managing IT support requests.

13. SYSTEM COMPONENTS

The project design consists of the following components:

Component | Purpose

Custom Ticket Table | Stores support tickets

Caller Field | Identifies the requester

Category Field | Stores major issue type

Subcategory Field | Stores specific issue type

Short Description | Provides issue summary

Description | Stores detailed issue information

Flow Designer | Automates ticket classification

Update Record | Updates ticket classification

Send Email | Sends notification to caller

14. DESIGN OBJECTIVES

The system design aims to achieve the following:

1. Provide a structured ticket management system.
2. Automate ticket classification.
3. Reduce manual classification effort.
4. Maintain Category and Subcategory dependency.
5. Provide consistent ticket classification.
6. Automatically update ticket records.
7. Notify the caller through email.
8. Support future expansion of classification rules.

15. DESIGN VALIDATION

The proposed design is checked against the requirements identified in Phase 2.

Requirement | Design Component

Automatic Classification | Flow Designer

Category Assignment | Update Record

Subcategory Assignment | Update Record

Category Dependency | Dependent Choice

Ticket Storage | Custom Ticket Table

Email Notification | Send Email

Structured Data | Defined Ticket Fields

Future Expansion | Additional Flow Conditions

This ensures that the proposed design addresses the identified project requirements.

16. PHASE 3 OUTCOME

At the end of the Project Design Phase, the complete system structure and workflow are defined.

The major design outcomes are:

1. Custom ticket table design.
2. Ticket field design.
3. Category design.
4. Subcategory design.
5. Category–Subcategory dependency design.
6. Ticket classification design.
7. Flow Designer workflow design.
8. Email notification design.
9. Complete automation workflow design.
10. User interface design.

17. CONCLUSION

The Project Design Phase defines the technical structure and workflow of the Auto Ticket Classification using Flow Designer project.

The system is designed with a custom ticket table, structured fields, dependent Category and Subcategory choices, Flow Designer automation, automatic ticket classification, and email notification.

The designed workflow provides a clear path from ticket creation to automatic classification, ticket update, and caller notification.

This design provides the foundation for the next phase, Project Planning Phase, where the implementation activities, sequence, resources, and project execution plan are organized.